INCH-POUND

MIL-PRF-32278B
4 June 2008
SUPERSEDING
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26 March 2008

PERFORMANCE SPECIFICATION

GAUGE, TACTICAL AUTOMATED TANK (TATG)

Fuels Operational Readiness Capability Equipment (FORCE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification describes a tactical automated tank gauging (TATG) system that is capable of accurately measuring the quantity of liquid petroleum in collapsible tactical fuel storage tanks, which are used throughout the United States Department of Defense (DoD) for storage of liquid petroleum products in tactical environments. The TATG should be composed of three components: measuring device, tank measurement receiver, and lightweight meter. The TATG should calculate the inventory of a collapsible tank with an accuracy rate of 1/8 inch , per American Petroleum Institute (API) ATG standards (chapter 3), and transmit via hard wire cable connection inventory data to a hand held device capable of communicating with industry standard Supervisory Control and Data Acquisition (SCADA) software through a download port, cradle, RS232, RS485 or Local Area Network (LAN) to the DOD Fuels Manager/Fuels Automated System (FAS).

1.2 Classification. Lightweight meter should be of the following type as specified:

Type 1 - 6 inch four ear female camlock inlet and 6 inch male camlock outlet and flow rates up to 1200 gallons per minute.

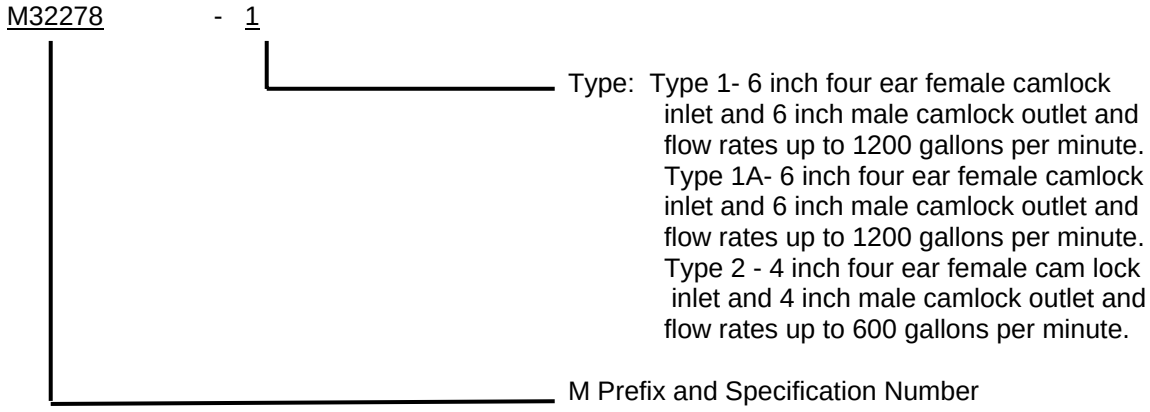
Type 2 - 4 inch four ear female camlock inlet and 4 inch male camlock outlet and flow rates up to 600 gallons per minute.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/GBEC, 480 Richard Ray Blvd. Suite 200, Robins AFB, GA 31098-1640. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1.3 Part or Identifying Number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor. The PIN to be used for TATG acquired to this specification is created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The listed documents are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections, recommended for additional information, or given as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.1.1 Government specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52530	Gasoline, Automotive, Unleaded
A-A-52557	Fuel Oil, Diesel

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4, JP-5, and JP-5/JP-8 ST
MIL-DTL-25524	Turbine Fuel, Aviation, Thermally Stable
MIL-DTL-83133	Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8), NATO F-35, and JP-8+100

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-276	Impregnation of Porous Metal Castings and Powered Metal Components
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-161	Identification Methods For Bulk Petroleum Products Systems Including Hydrocarbon Missile Fuels

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-808	Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

DEPARTMENT OF DEFENSE DIRECTIVE

DoD 4140.25	Management Policy for Energy Commodities and Related Services

(Copies of this document is available online at http://www.dtic.mil/whs/directives/corres/dir.html .)

2.1.2 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents, which are adopted, are those cited in the solicitation or contract.

American Petroleum Institute

API RP 2003	Protection against Ignitions Arising out of Static, Lightning, and Stray Currents

American Petroleum Institute, Manual Petroleum Measurement Standards, Chapter 3 Section 1B—Standard Practice for Level Measurement of Liquid Hydrocarbons in Stationary Tanks by Automatic Tank Gauging

American Petroleum Institute Manual Petroleum Measurement Standards, Chapter 7, Temperature Determination

American Petroleum Institute Manual Petroleum Measurement Standards, Chapter 9, Density Determination

(Copies of these documents are available online at http://www.api.org/ or from the American Petroleum Institute, 1220 L Street, NW, Washington, D.C. 20005, phone 202/682-8000 or FAX 202/682-8030.)

ASTM INTERNATIONAL

ASTM-D1655	Standard Specification for Aviation Turbine Fuels
ASTM-D910	Standard Specification for Aviation Gasolines

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, West Conshohocken PA 19428.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	National Electric Code

(Copies of this document is available online at http://www.nfpa.org/ or from the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

2.2 Order of precedence. In the event there is a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS.

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.

3.2 Materials. All components and all materials shall be selected based on the defined purpose for the specified environmental extremes stated herein. Materials shall be certifiable for the application. Components shall not be used outside their published ratings. All components and materials that normally come into contact with the fuel during operation (including gaskets, fittings, valves, adhesives) shall be proven fuel-resistant materials, shall not degrade, or be affected by fuels cited in this specification. The TATG shall be fabricated from strong, durable and lightweight materials to the maximum extent possible to reduce overall unit weight. The TATG shall be fabricated from compatible materials, inherently corrosion-resistant, or treated to provide protection against corrosion and deterioration that may be encountered.

3.2.1 Prohibited materials. Magnesium alloys, wood products, polyvinyl chloride (except for wiring covering), polyester, room temperature vulcanizing materials (yielding acetic acid), or asbestos shall not be used in any component of this tactical automated tank gauging system.

3.2.2 Metals. If the design of the TATG uses the collapsible fabric coated tank vent stack assembly for attachment, metals used in the TATG must be compatible with aluminum. All metallic components shall be protected from corrosion. All metallic non-aluminum fuel-handling components shall be corrosion and fuel resistant. All aluminum fuel-handling components shall be anodized or chemically conversion-coated without deleterious effects on the function, performance, or availability of these components. Copper-based materials shall not come in direct contact with fuel, except within components certified for use with the fuels listed herein. Guidance for dissimilar metals and means of protection is provided in MIL-STD-889.

3.2.3 Impregnation of castings. All castings shall be pressure tested. Aluminum castings may be impregnated in accordance with MIL-STD-276 if required to prevent weeping.

3.2.4 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable.

3.2.5 Elastomers. Elastomeric materials used shall be compatible with fuels conforming to ASTM-D910, A-A-52557, MIL-DTL-5624, A-A-52530, ASTM-D1655, and MIL-DTL-83133 and shall withstand -60° to +160°F temperatures.

3.2.6 Protective treatment. Materials used in the construction of the TATG shall be protected against deterioration likely to occur due to climatic or environmental conditions during operation. However, that protection shall in no way prevent compliance with the performance requirements of this specification. The manufacturer shall avoid using any protective coating that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions. Guidance is provided in MIL-HDBK-808.

3.2.7 Servicing provisions. Special tools shall not be required for servicing. The design shall minimize the number of parts to be dissembled and reassembled and to minimize the different types of hand tools required for maintenance.

3.2.8 Fool-proofing. Components shall be designed, by shape or by mounting pattern, to prevent improper installation.

3.2.9 Foreign object damage (FOD). Any metal cap, plug, pin, or plate that must be removed for inspection, service or operation, shall be retained by wire rope lanyard or chain to prevent it from becoming separated from the TATG. “Dog tag” style beaded chains shall not be used.

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3.2.10 Bonding. All metal components shall be bonded in accordance with the guidelines of API RP 2003.

3.2.11 Prevention of static electricity. Nonmetallic components shall be certified to a resistance of 10 ohms or less to ground.

3.3. Recycled, reclaimed, or recovered materials. Recovered materials shall be used for fabrication of the TATG system to the maximum extent possible, without affecting the intended use. Used or rebuilt parts shall not be defined as recovered materials.

3.4 Design. The TATG shall conform to all Federal laws and regulations applicable for this type equipment in effect on date of issuance of solicitation. The system shall be designed to meet the following requirements:

a. Major components must be capable of being readily installed, operated and removed by personnel or maintenance technician wearing chemical protective or arctic-weather gloves.

b. Rapid installation, maintenance, and repair of equipment.

c. Parts shall not work loose while in service and shall be inherently capable of withstanding the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5 Safety. A system safety program that complies with the requirements of MIL-STD-882 shall be established. A System Safety Analysis (SSA) shall establish the risk levels associated with the fueling systems. A Subsystem Hazard Analysis (SSHA) shall include, in industry terms, a Failure Mode Effects and Criticality Analysis (FMECA). Additionally, in case of fabric coated tank rupture, the measurement device must be constructed in such a manner that complete immersion in fuel does not create a hazard.

3.5.1 Inadvertent ignition. System shall be designed to preclude inadvertent ignition and to perform effectively, during or after exposure to the operational electromagnetic environment and while operating electronic controls in a fueling environment

3.6 Components.

3.6.1 Measurement device (MIL-PRF-32278 M/M DEV-1). Each Type 1 TATG system shall contain ten (10) each measurement devices, each Type 1A TATG system shall contain twenty (20) each measurement devices and each Type 2 TATG system shall contain six (6) each measurement devices. The measurement device shall be designed and constructed to accurately measure the level of liquid petroleum in collapsible coated fabric tactical fuel storage tanks, as are used throughout DoD. The accuracy of the measurement device shall be 1/8 inch. The measurement device shall measure the inventory level of a tank, convert the measure to a corresponding net volume in gallons at 60°F, and transmit via hardwire communication, inventory data to the tank measurement receiver. The hardwire extending from each measuring device to the hand held receiver should not be less than 70 feet and not more than 100 feet in length. The measurement device shall withstand conditions incidental to operation, shipping, and storage, at the environmental extremes specified herein. The system components that come into contact with fuel shall be compatible with fuels conforming to ASTM-D1655, MIL-DTL-5624, A-A-52530, MIL-T-25524, ASTM-D910 and MIL-DTL-83133. The only maintenance action necessary shall be recharging/replacement of batteries when necessary. The battery must last 6 months without replacement. The construction of the tank measurement device shall be such that battery replacement can be accomplished in 1 hour or less. If the design of the tank measurement device requires connection to the tank vent, the tank measurement device must allow tank vapor venting to atmosphere without degrading the current capability. The measuring device may be powered through the receiver cradle via communication cables.

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3.6.2 Tank measurement receiver (MIL-PRF-32278 TK M/M REC-1). Each Type 1 and 1A TATG system shall contain two (2) each measurement receivers and each Type 2 TATG system shall contain one (1) each measurement receiver. The tank measurement receiver is a hand held device capable of retrieving, displaying, and storing corrected fuel inventories for collapsible tanks and communicating with industry standard SCADA software through a download port, cradle, RS232, RS485 or LAN connection. The tank measurement receiver shall be capable of retrieving, displaying, and storing inventory data from up to 100 tank measurement devices. The receiver shall facilitate easy maintenance and removal of components with minimum disassembly. The only maintenance action necessary shall be recharging/replacement of batteries when necessary. The construction of the tank measurement receiver shall be such that battery replacement can be accomplished in one hour or less.

3.6.3 Training and re-training of the measurement device. Upon first use, the tank measurement device shall be attached to the collapsible tank, and a metered fill shall occur. The tank measurement device shall communicate (pulse) with the lightweight meter during the initial fill and capture data points in gallons corresponding to 1/8 inch increments up to a max height of 72 inches. During the initial training, in the case of a protracted fill operation, the tank measurement device shall retain these data points in memory until training is completed. The tank measurement device shall incorporate a feature that develops a tank-strapping chart in feet and inches with corresponding data points in gallons and with temperature/density probes shall correct fuel to 60°F. Once trained, the tank measurement device shall hold the data points in memory until erased by fuels personnel. The tank measurement device shall incorporate a feature that allows the device to be reprogrammed/reused on collapsible tanks of the same or different sizes.

3.6.4 Lightweight meter (Type 1: MIL-PRF-32278 MET L/W-6, Type 2: MIL-PRF-32278 MET L/W-4). Each Type 1 and 1A TATG system shall contain two (2) each lightweight meters and each Type 2 TATG system shall contain one (1) each lightweight meter. The meter accuracy shall be in accordance with NIST Handbook 44 Section 3.30, table T.2 Accuracy Class 0.3. The meter shall be capable of communicating (transmitting data) with the measurement device in US gallons. The lightweight meter shall be rated at a minimum of 150 psig. Each Type 1 light weight meter shall include one each reducers, 6 inch camlock coupler to 4 inch adapter and one 4 inch camlock coupler to 6 inch adaptor. The lightweight meter shall be easily transported and installed by one person.

3.6.5 Batteries. Batteries shall be commercially available lithium batteries capable of meeting the temperature extremes required in this specification. Batteries shall be interchangeable with non lithium standard commercially available batteries for emergency use. TATG batteries shall last a minimum of 6 months of use without replacement. Where practical maximize use of interchangeable batteries for the TATG components.

3.7 Construction.

3.7.1 Threaded fasteners. Threaded fasteners shall include prevention for the loss of torque, while allowing for disassembly. Internal threads of aluminum components shall withstand not less than 20 fastener installation cycles to required torque. Adhesive-backed wiring supports shall not be used. The use of studs shall be minimized to prevent stripping of holes, except in low-pressure areas where bolt and nut arrangement cannot be readily applied. No less than three threads shall protrude from each nut fastener. Where studs are used, a suitable thread-locking agent shall be applied to prevent loosening.

3.7.2 Permanent fasteners. Permanently fastened, overlapping surfaces shall be sealed. Rivet selection shall be for a minimum joint strength. Rivet heading shall be consistent with published data for a matching application. Riveting operations shall be carefully performed to ensure tight and satisfactory heading of rivets.

3.7.3 Wiring. The electrical wiring shall meet provisions of Class 1, Division 1 locations as defined by the National Electrical Code (NFPA 70). Wiring shall be installed in accordance with current industry practices for refueling tank and refueling vehicle applications. Wiring shall adhere to NFPA 70 for wiring in hazardous classified locations. All wiring shall be enclosed in rigid metal conduit or flexible conduit

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covered with polyvinyl chloride. All switches and electrical devices shall be hermetically sealed unless in a vapor-tight enclosure. The entire electrical system and all components shall be designed to withstand an 11-psi positive or negative pressure without leaking. The electrical wire shall be a minimum of 18-gauge wire with moisture-resistant thermoplastic insulation.

3.7.4 Welding. Cyclic duty welding shall conform to ASME-B31.3. Weld beads shall be thoroughly cleaned, and spattering shall be removed.

3.8 Hardware.

3.8.1 Screw assemblies. Assembly screws and bolts shall be torqued within commercial limits, so as not to loosen during transport or operation. Assembly screws and bolts shall be removable by one person using only hand tools.

3.9 Component accessibility. The TATG unit shall be designed to enable all components to be readily removed, serviced, and operated. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates that must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners.

3.10 Reliability and Maintainability.

3.10.1 Reliability. The TATG system (measuring device and receiver) shall be designed and constructed for long life by minimizing battery use. The measuring device shall incorporate a feature that allows the unit to remain in standby mode until interrogated. The measuring device, meter, and receiver shall operate for a minimum of six months without battery replacement.

3.10.2 Maintainability. The TATG shall be designed to facilitate ease of maintenance and removal of components with minimum disassembly. The only maintenance action necessary shall be replacing batteries in the tank measurement transmitter. The TATG shall be constructed to allow any battery to be changed in one hour or less.

3.11 Performance requirements. If the tank measuring device is to rest on any portion of the collapsible tank, or on or in the tank vent stack, it must be lightweight enough to not cause undue stress and must not interfere with the tank's ability to vent fuel vapor. The tank measuring device shall not be designed in a manner that makes battery changes unduly difficult for the operator, and shall incorporate hardware that prevents memory loss when changing batteries. TATG system shall meet the following characteristics:

a. Be capable of accuracy as set forth in the DOD 4140.25M for measurement of liquid petroleum products that have specific gravity range of between 0.65 and 0.85.

b. The TATG system shall be certified to operate in a Class 1, Division 1 environment.

c. The TATG system shall not produce any energy (RF emission) that would in any way be capable of igniting the vapors produced by the product within the tanks.

d. The system shall not emit any energy of any kind that could be detected beyond 2,000 feet in any direction.

e. The TATG system shall not use any external power source as the primary source of operating power.

f. The TATG system shall not produce any energy that will be harmful to personnel.

g. The TATG shall be capable of producing output that shall be compatible with industry standard

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SCADA software.

h. The tank measurement receiver shall be able to support a Microsoft® Windows CE- or equivalent based program that can display fuel inventory level, (in feet and 1/8 inch increments), unadjusted quantity in gallons, adjusted quantity in gallons,(corrected to 60 degrees), temperature and density for each tank. The receiver shall be easily read in full sunlight or nightime conditions. Components shall be capable of full operation for no less than six months without replenishment or repair. The receiver shall be rugged and durable.

3.12. Environmental requirements.

3.12.1 Electromagnetic interference (EMI). The TATG shall be in accordance with the electromagnetic interference (EMI) requirements of MIL-STD-461.

3.12.2 Storage temperatures. The TATG shall not be damaged by storage in ambient temperatures ranging from -60° to +160°F for not less than 50 hours (when tested in accordance with 4.5.4.1 and 4.5.4.2).

3.12.3 Operating temperatures. The TATG shall be capable of performing as specified herein in ambient temperatures between -20° and +140°F (when tested in accordance with 4.5.4.1 and 4.5.4.2).

3.12.4 Humidity. Operation or storage in any relative humidity up to and including 100% shall not damage the TATG; including conditions where condensation takes place in the form of water and frost (when tested in accordance with 4.5.4.3).

3.12.5 Sand and dust. The TATG shall not be damaged by operation or storage in atmospheres containing sand and dust particles as encountered in desert areas (when tested in accordance with 4.5.4.4).

3.12.6 Rain. The TATG shall not be damaged by operation or storage in rainfall as encountered in any locale. The tank measurement system shall not have areas that trap and retain water from rainfalls (when tested in accordance with 4.5.4.5).

3.12.7 Fungus. The TATG shall not be damaged by exposure to moist fungus growth such as encountered in tropical and subtropical climates (when tested in accordance with 4.5.4.6).

3.12.8 Salt-Fog atmosphere. The TATG shall not be damaged by operation or storage in atmospheres containing salt-laden moisture such as encountered near bodies of salt water and in transportation on shipboard (when tested in accordance with 4.5.4.7).

3.13 Human factors. The TATG shall be designed and constructed to allow general maintenance, assembly, and disassembly to be performed with common hand tools in accordance with MIL-STD-1472.

3.14 Hand tools. The TATG system shall be designed and constructed to allow general maintenance, assembly, and disassembly to be performed with common hand tools.

3.15 Seals. The TATG shall be sealed so portions of the apparatus coming in contact with the fuel will not contaminate the fuel or be damaged by the fuel.

3.16 Safety. A system safety program that complies with the requirements of MIL-STD-882 shall be established. A System Safety Analysis (SSA) shall establish the risk levels associated with the fueling systems. A Subsystem Hazard Analysis (SSHA) shall include, in industry terms, a Failure Mode Effects and Criticality Analysis (FMECA). The SSHA shall include methods for controlling any identified hazard. All exposed parts that are subject to high operating temperatures shall be insulated, fully enclosed, or guarded. All moving parts that are hazardous to operating or maintenance personnel shall be fully

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enclosed or guarded. Protective devices shall not impair the operating functions. No exposed surface shall be capable of generating a temperature of 750°F.

3.17 Identification markings. Instruction and identification data shall be permanently and legibly imprinted upon data plates or upon the device itself in accordance with MIL-STD-130.

3.18 Identification. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 and MIL-STD-161 for specific fuel-handling components. Separate Unique Item Identification requirements are required under DFARS 252.211-7003, as outlined in the basic contract. Instruction and identification data shall be permanently and legibly etched upon metal plates. The plates shall be aluminum alloy with a minimum thickness of 18 gauge (0.04 inch). Component identification plates shall be aluminum alloy with a minimum thickness of 0.032 inch. All plates shall be anodized or etched to produce raised natural metallic colored markings with a dark color background.

3.19 Workmanship. The TATG, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alignment of parts; and tightness of assembly screws and bolts.

4. VERIFICATION

4.1 Classification of inspections. The inspection and testing of the TATG shall be classified as follows:

a. First-article inspection. First article inspection shall be performed on the first complete Type 1 TATG system and complete Type 2 TATG system. This inspection shall include reference tests listed in 4.3

b. Conformance inspection. Conformance inspections shall be performed on each TATG provided by the manufacturer. The conformance inspection shall include all examinations and tests contained in 4.4.

4.2 Test conditions. All data shall be corrected to sea-level conditions, with a barometric pressure 29.92 inches of mercury, at 60°F. Unless otherwise specified herein or in the detailed component specifications, the test fluid shall be Aviation Turbine Fuel, Grade JP-8 conforming to MIL-DTL-83133 or Jet A-1 commercial aviation fuel.

4.3 First-article inspection testing.

4.3.1 Test sample. Unless otherwise specified, one Type 1 TATG system and one Type 2 TATG system shall be subjected to the test methods specified under paragraph 4.5.

4.3.2 First article conformance inspection. Each Type 1 TATG system and Type 2 TATG system shall be subjected to the following tests described below:

a. Examination of product	4.5.1
b. Mechanical operation	4.5.2
c. Electromagnetic Interference (EMI)	4.5.3
d. Environmental test	4.5.4
e. Endurance test	4.5.5

4.4 Conformance inspection. Each TATG shall be subjected to the following tests described below:

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a. Examination of product 4.5.1

The manufacturer shall provide certification, verifying compliance with the tests noted above.

4.5 Test methods for verifying requirements. The TATG shall be subjected to the tests shown below. During these tests the following data shall be recorded and a statement of compliance shall be provided along with the certifying official's name and date for each test.

a. Date

b. Time started / finished

c. Ambient temp

d. Fuel temp (supply)

e. Unadjusted Fuel Density

f. Adjusted Fuel Density

g. TATG tank measurement device Fuel Volume (gallons)

h. TATG tank measurement device Corrected Fuel Volume (gallons)

i. Fuel level (feet and inches)

j. Fuel gallons metered in (independent calibrated meter for verification)

k. Fuel Gallons metered out (independent calibrated meter for verification)

l. Lightweight meter K-factor

m. Type product used

4.5.1 Examination of product. The TATG shall be inspected to determine compliance with the requirements specified herein and shall be examined for defects. The presence of one or more defects shall be cause for rejection. Particular attention shall be given to compliance with design requirements on preproduction assemblies; and to materials, workmanship, dimensions, protective finishes and their applications, welding, riveting, and markings on the TATG. The TATG shall be examined to ensure that protection is provided for devices/components that could otherwise easily be inadvertently damaged. Reasons for rejection include but are not limited to the following:

a. TATG equipment not as specified in contract or order.

b. Materials not resistant to corrosion or deterioration.

c. Dissimilar metals not protected against galvanic corrosion.

d. Level of preservation, packing and marking not as specified in the contract.

4.5.2 Mechanical operation test. The mechanical operation test shall be conducted on each measurement device to independently verify its accuracy during a fill operation. During this test, an independent meter (certified accurate) shall be used in addition to the TATG lightweight meter during the fill operation to verify TATG captured data points. The fill shall be stopped at 1/8 inch increments and the measurement device queried and reading compared to independent meter readings to verify accuracy of the lightweight

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meter. Verification of density and temperature measurement accuracy will also be accomplished during this phase in accordance with paragraph 2.1.2 Non-Government Publications. During this testing, all components that come in contact with fuel shall be tested and the temperature and density measurements recorded on all fuels listed in section 3.2.5. Certificate of conformance (COC) shall be provided. Mechanical operation testing may be conducted in any size container that will provide sufficient test result data to satisfy the requirements stated herein.

4.5.3 Electromagnetic interference (EMI) test. The TATG shall be tested for Radiated Emission in accordance with RE102 of MIL-STD-461 and Radiated Susceptibility in accordance with RS103 of MIL-STD-461. The TATG shall be tested for Conducted Susceptibility in accordance with CS114, CS115 and CS116 of MIL-STD-461.

4.5.4 Environmental tests. The following environmental tests shall be conducted on the TATG in accordance with the specified methods of MIL-STD-810. The engineering evaluation shall consist of a written, detailed analysis of the TATG and all its parts and components to satisfactorily accomplish the required environmental conditions. Environmental testing may be conducted in any size collapsible tank that will provide sufficient test result data to satisfy the requirements stated herein. Fluids other than stated in 4.2 may be used during environmental testing.

4.5.4.1 High temperature test. The TATG shall be subjected to a temperature of 160°F for 24 hours. No adverse effects that would affect operation at normal temperature shall result from this exposure. The chamber temperature shall be reduced to 140°F, and the tests specified in 4.5.1 and 4.5.2 shall be conducted. The TATG shall be operated for 3 hours. The TATG shall operate satisfactorily and perform its intended functions throughout this test. Leakage, damage, evidence of thermal degradation, or impending failure of any portion of the TATG shall be cause for test failure. Testing shall be in accordance with MIL-STD-810, Method 501.4.

4.5.4.2 Low temperature test. The TATG shall be subjected to a temperature of –40°F for a period of 24 hours. No adverse effects that would affect operation at normal temperature shall occur as a result of this exposure. The chamber temperature shall be increased to -20°F and the tests specified in 4.5.1 and 4.5.2 shall be conducted. The TATG shall be operated 3 hours. The TATG shall operate satisfactorily and perform its intended functions throughout this test. Leakage, damage, evidence of thermal degradation, or impending failure of any portion of the TATG shall be cause for test failure. Testing shall be in accordance with MIL-STD-810, Method 502.4.

4.5.4.3 Humidity. The TATG shall be subject to natural cyclic high humidity conditions that randomly vary the temperature between 79° and 95°F with the relative humidity reaching between 85 and 95% over a 24-hour period. Immediately following the humidity cycling, without drying or allowing the unit to dry, perform tests as specified in 4.5.1 and 4.5.2. Leakage, damage, evidence of degradation, or impending failure of any portion of the TATG shall be cause for test failure. Testing shall be in accordance with MIL-STD-810, Method 507.4.

4.5.4.4 Sand and dust. The TATG shall be subject to a test chamber temperature minimum 73°F with a relative humidity (maintained throughout the test) maximum 30%. Subject the TATG to a blowing dust concentration (97 to 99 % by weight) of silicon dioxide with particle size ranging from l00 to 325 microns at air velocities ranging from 300 to 1750 feet/minute for minimum 6 hours. Stop the test and allow the test article to return to ambient conditions. Remove accumulated dust from the test article by brushing, wiping, or shaking. Do not remove dust by either air blast or vacuuming. Take care not to introduce additional dust into the test article. Perform the test as specified in 4.5.1 and 4.5.2 with no degradation in appearance or performance. Repeat the test using sand concentrations of 2.2 to +/-0.5 g/m. Perform the tests as specified in 4.5.1 and 4.5.2 with no degradation in appearance or performance. Leakage, damage, evidence of thermal degradation, or impending failure of any portion of the TATG shall be cause for test failure. Testing shall be in accordance with MIL-STD-810, Method 510.4.

4.5.4.5 Rain. The TATG shall be subject to an instantaneous rainfall rate equivalent to 0.80 mm/minute, a minimum rate of 10 cm/hr. The rainfall intensities shall be accompanied by winds of 18 m/s (40 miles

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per hour). The rain shall be produced by a water distribution device of such design that the water is emitted in the form of droplets having a diameter range between 0.5 and 4.5 millimeters. The rain shall be dispersed completely over the test item for 15 minutes. The wind at the velocity specified in the test plan shall be maintained for minimum 30 minutes. Rotate the test item to expose the rain source to the other side of the test item that could be exposed to blown rain in the deployment cycle. The rain test equipment shall not allow any amount of free water to penetrate the test item. Measure and document the volume of water (if any) inside the protected areas of the TATG. Leakage, damage, evidence of thermal degradation, or impending failure of any portion of the TATG shall be cause for test failure. Testing shall be in accordance with MIL-STD-810, Method 506.4.

4.5.4.6 Fungus. The TATG shall be subject to fungi and possibly abundant growth colonization of 50% or more of the area of the control item after 14 and 28 days. The unit shall not be cleaned for the first article for 72 hours prior to the fungus test. The unit shall not be contaminated before or during the test by handling. Perform the tests as specified in 4.5.1 and 4.5.2 with no degradation in appearance or performance. Leakage, damage, evidence of thermal degradation, or impending failure of any portion of the TATG shall be cause for test failure. Testing shall be in accordance with MIL-STD-810, Method 508.5.

4.5.4.7 Salt-Fog. Using spray apparatus, piping and collector reservoir materials that are non-reactive to salt spray, subject the filter separator to a continuously atomized, finely divided, wet, dense salt spray mixture (5% sodium chloride and 95% water) at a constant 95°F with minimum air circulation. Do not use immersion heaters in the exposure area. Uniformly distribute the salt fog over the TATG at a fallout rate between 0.01 and 0.1 ounces per ½ hour for a minimum of 48 hours. After the test, store the unit at ambient atmosphere minimum 48 hours for drying. Perform the tests as specified in 4.5.1 and 4.5.2. Leakage, damage, evidence of thermal degradation, or impending failure of any portion of the TATG shall be cause for test failure. Testing shall be in accordance with MIL-STD-810, Method 509.4.

4.5.5 Endurance test. The TATG system shall be subjected to the following endurance test on the first article only at the government furnished test site. The TATG system shall be used to determine inventories on all collapsible tanks at the test location for a period up to but not more than 30 days. The TATG system shall show no indication of impending failure or degradation of components due to this test.

a. A control measurement device shall be attached to at least one 50,000-gallon collapsible tank and an independently metered fill shall occur. Contractor shall provide two each 4-inch, and one each 6-inch, certified accurate, temperature compensating meters and all connecting hardware. The tank measurement device shall be trained by the lightweight meter (ref section 3.6.3). The lightweight meter shall be removed and the tank shall then be drained in 2,000-6,000 gallon independent metered increments, and the unit queried at each increment to verify accuracy. The test shall consist of a minimum of three cycles and a maximum of six cycles (see 6.6.1). The tank measurement device shall be zeroed and retrained every two cycles with the lightweight meter. Additionally, the device from the 50,000-gallon collapsible tank shall be used on a 210,000-gallon collapsible tank. The tank measurement device shall be trained by the lightweight meter (ref section 3.6.3). The lightweight meter shall be removed and the tank shall then be drained in 2,000-6,000 gallon independent metered increments, and the unit queried at each 2,000-6,000 gallon increment to verify accuracy. The test shall consist of a minimum of three cycles and a maximum of six cycles (see 6.6.1). The tank measurement device shall be zeroed and retrained every two cycles with the lightweight meter.

b. The test above shall be repeated on a control measurement device on a 210,000-gallon collapsible tank. One measurement device from a 210,000 gallon tank shall be used on a 50,000 gallon tank as defined above.

c. In addition to the test identified above, the remainder of the measurement devices shall be installed on 50,000 and 210,000 gallon tanks. The measurement devices shall be connected to the light-weight meter for training on the initial fill. Fuel receipts, transfers and issues shall

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be tracked and compared to metered operations for the duration of the test to validate accuracy and record inventories.

Failure of any TATG system components during the duration of the test shall be cause for test failure.

4.5.6. Tank measurement receiver test. During the endurance test (ref section 4.5.5 a and b) each tank measurement receiver shall be downloaded at 5 random intervals to verify proper operation, data accuracy, and transferability into industry standard SCADA software. Additionally, the remainder of the measurement devices shall be downloaded as required for inventory reporting purposes throughout the duration of the test.

4.5.7 Reliability and maintainability. The following information shall be included as an attachment accompanying the first article inspection report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made on production items, or proposed to be made and the predicted effectiveness of these corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The TATG is intended to be a tactical fuel management system for the DoD and is intended to accurately measure the quantities of liquid fuel product within each collapsible storage tank.

6.2 Acquisition requirements. Procurement documents will specify the following:

a. Title, number, and date of this specification.

b. Type system required should be specified (see 1.1 & 1.2).

6.3 Qualification note. The contractor will certify and maintain substantiating evidence that the product offered meets all characteristics of this specification and that the product conforms to the producer's own

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drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter, as may be otherwise provided for under the provisions of the contract.

6.4 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, provided they fall within the tolerances specified in their corresponding inch/pound units.

6.5 Contractor Endurance Test Location. The government will provide access to a testing location for the endurance testing portion of the first article test. Access to all tanks, pumps and fuel will be provided by the government to complete testing. When the testing is performed at an overseas government deployed location, the government will provide lodging at no charge and access to messing facilities at a service charge fee per meal at the test site. Government will provide contractors transportation to and from airport at the beginning and conclusion of the test and provide daily transportation to test sites. Government will complete all actions necessary to provide access to test location. Contractor will be responsible for providing information for clearances and follow DoD guidelines for entry into the test location. All travel to prepare for departure and to test site will be the responsibility of the contractor. Portions of first article tests that can not be completed at the test site e.g. cold weather and salt fog etc. will be the responsibility of the contractor.

6.6 Definitions

6.6.1 Cycle. Cycle is defined as a tank being filled to maximum capacity and then drained empty.

6.7 Subject term (key word) listing.

Collapsible
Light weight meter
Measuring device
Measurement receiver

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force – 99

Preparing activity:
Air Force – 84

Agent:
Air Force – 99

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